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Construction of hierarchical porous carbons with Ph-POSS based polymers for highly efficient adsorption of polycyclic aromatic hydrocarbons

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ABSTRACT

Fluid catalytic cracking (FCC) diesel provides most polycyclic aromatic hydrocarbons (PAHs) in diesel. The removal of PAHs by adsorption and separation method is an efficient way to obtained purified diesel. As a kind of green economy adsorbent, porous carbon materials (PCMs) have been widely used in the adsorption of gases, aromatics and organic pollutions. Here, by direct carbonization, the PCMs are prepared from the novel Ph-POSS based porous organic polymers (POPs) which possessed high BET surface areas (~1000 m^2/g). The PCMs possess abundant microporosity and O atoms in the skeleton. The PCMs exhibit high PAHs adsorption capacity because of the hierarchical micropore size, high degree of graphitization and abundant O atoms. The strategy of construction PCMs with hierarchical micropore and heteroatoms provides an effective method to remove PAHs in the FCC diesel.

1. Introduction

With the development of global industrialization, the demand for fluid catalytic cracking (FCC) diesel is increasing.[1,2] The percentage of polycyclic aromatic hydrocarbons (PAHs) accounts for a high proportion in FCC diesel which have been widely concerned by researchers as serious environmental hazards and biological health hazards.[3,4] However, PAHs are also important industrial chemicals in synthesis of dyes and pesticides.[5,6] Therefore, it is vital to find effective methods to reduce the content of PAHs in the FCC diesel. Adsorption is a widely studied method to remove PAHs because of the easy processing, mild operating conditions, and environmentally friendly.[7] The design and synthesis of adsorbents with high selective adsorption ability and high capacity are first consideration in adsorption method.[8].

The porous materials are ideal adsorbents because of their designable structures and pore size.[9,10] Some different porous materials, such as zeolites, [11] metal organic frameworks (MOFs), [12] porous organic polymers (POPs)[13] and the derived porous carbon materials (PCMs)[14] have been well studied in the adsorption of gases, [15] metal ions [16] and organic pollutions [17]. POPs are ideal precursors to obtain carbonized PCMs because of the tunable pore size, good stability, and high porosity.[18–20] Compared the synthesis method of PCMs with the template, the direct carbonization of POPs avoids the use of acid or base and the operation processes are economic and time saving.[21,22] The excellent adsorbents usually possess high surface area, proper pore size and volume, and the abundant interaction sites with adsorbates.[23,24] To design the adsorbents of PAHs, the molecular size and rich π electrons of PAHs are need to be taken into full consideration. Due to the inherent π system of PAHs, the porous materials with open channels, π system and sites which can form interaction forces with PAHs are potential candidates.[25,26] Especially, the PCMs with heteroatoms (eg. N, O) can introduce extra p-π conjugated system with PAHs to enhance the adsorption capacity.[27] Octaphenyl polyhedral oligomeric silsesquioxanes (**Ph-POSS**), as a rigid organic–inorganic hybrid monomer with high steric hindrance, had been used in the synthesis of POPs.[28–30] The high steric hindrance of **Ph-POSS** can reduce the packing of polymers effectively and widely used in gas sorption, water treatment, catalysis and sensors.[31–33].

In this study, we synthesized three novel POPs (**PPOSS-1**, **2**, **3**) by Friedel-Crafts alkylation based on **Ph-POSS** under mild conditions. **PPOSS-1**, **2**, **3** possessed high BET surface area (~1000 m^2/g) and narrow microporosity. The derived PCMs were obtained by

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carbonization of the polymers at different high temperature directly. After carbonization, the porosity was kept and the micropore size was increased. The breakage of Si-O-Si in PCMs which led by carbonization exposed more O atoms in the skeleton. Due to the suitable pore size, sufficient π-π interaction and p-π conjugation, the PCMs exhibited high PAHs adsorption capacity (20.58 mg/g, naphthalene; 58.92 mg/g, anthracene; 17.30 mg/g, acenaphthene; 26.00 mg/g, acenaphthylene) in the simulated diesel. The prepared PCMs as PAHs adsorbents provide an efficient way to separate PAHs/long-chain alkane and purify the FCC diesel.

2. Experimental sections

2.1. Materials

4,4′-Bis(chloromethyl)-1,1′-biphenyl (**BCMBP**), 1,4-bis(chloromethyl)benzene (**BCMB**), 1,3,5-tris (bromomethyl) benzene (**TBMB**) and octaphenyloctasilsesquioxane (**Ph-POSS**) were obtained from Macklin Inc. Anhydrous aluminum chloride ($AlCl_3$), hydrochloric acid (HCl), acetone, tetrahydrofuran, methanol and dichloromethane were purchased from Aladdin Biochemical Technology Co. Ltd. All chemicals and solvents were used as received without any purification.

2.2. Characterization

The Fourier transform infrared (FTIR) spectra were collected on a Nicolet Nexus 470 FTIR spectrophotometer (Thermo Electron Corporation, USA) with the KBr disk method. The field-emission scanning electron microscope (SEM, JSM-7001F, JEOL, Japan) equipped with energy dispersive spectrometer (EDS) and transmission electron microscopy (TEM, JEM-2100, JEOL) were used to explore the morphology of the materials. The Micromeritics ASAP 2020 HD88 surface area and porosity analyzer (Micromeritics Instrument Corporation, USA) were used to record the N_2 adsorption and desorption isotherms, and the pore size distributions were obtained by the nonlocal density functional theory (NLDFT) calculation. X-ray powder diffraction (XRD) was performed at the scanning rate of 5° min^{-1} from 5° – 50° on a Shimadzu XRD-6100 Lab X-ray diffractometer with Cu Kα radiation. The X-ray photon spectroscopy (XPS) was performed at a VG MultiLab 2000 spectrometer (VG Instruments, UK) with an Mg Kα X-ray source. Thermogravimetric (TG) analysis was measured on STA 449C (NETZSCH Corporation, Germany) at a heating rate of 10 °C min^{-1} from 20 °C – 800 °C under air.

2.3. Synthesis of POPs

2.3.1. Synthesis of PPOSS-1

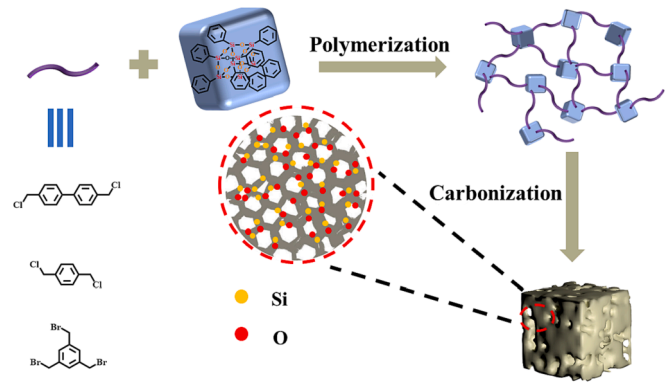
In a typical synthesis process, **BCMBP** (0.5 mmol, 627.85 mg), **Ph-POSS** (0.1 mmol, 516.75 mg) and $AlCl_3$ (1 mmol, 1.21 g) were added into the 250 ml two-neck round bottom flask. The system was degassed and then refilled with nitrogen with three times. CH_2Cl_2 (150 ml) was injected into the system and refluxed for 2 days. The obtained product was collected with filtration and washed with 1 M HCl, methanol, tetrahydrofuran and dichloromethane to remove unreacted monomers, catalysts and oligomers. The yellow–brown powders were obtained and dried 120 °C for 8 h (**PPOSS-1**, yield: 92 %).

2.3.2. Synthesis of PPOSS-2 and PPOSS-3

The synthesis procedure is same except that **BCMB** (0.5 mmol, 437.65 mg) was used to get **PPOSS-2** (yield: 87 %) and **TBMB** (0.5 mmol, 713.75 mg) was used to get **PPOSS-3** (yield: 73 %).

2.4. Synthesis of PCMs

The obtained **PPOSS** polymer (100 mg) was placed in a porcelain ark and then put into tubular furnace. The tube was flowed with N_2 for half hour before heating. Then the system was heated to setting temperature with a heating rate of 5 °C min^{-1}. After keeping for 2 h, it was cooled down to RT with a rate of 5 °C min^{-1}. A series of porous carbon materials were obtained at different carbonization temperatures (700 ℃, 800 ℃, 900 ℃, 1000 ℃). The product was washed with methanol and then dried at 80 °C for 8 h (Scheme 1).



Scheme 1. The schematic diagram of synthesis of **PPOSS** polymers and **PCMs**.

2.5. Adsorption of PAHs

Firstly, PAHs with the concentration of 200 ppm were prepared in *n*-cetane. Then 5 mg polymers were added to the 5 ml solution directly and the system was stirred for 2 h at RT. After reaching adsorption equilibrium, the polymers were removed by filtration. The concentration of PAHs was determined with GC.

2.6. Recycle test of POPs for PAHs adsorption

As the description in section 2.5, after adsorption equilibrium, the polymer was collected by filtration and regenerated by washing with CH_2Cl_2. The PAHs adsorption was carried out for 6 cycles.

3. Results and discussion

The structure of the polymers that synthesized with inorganic–organic hybrid **Ph-POSS** and different linkers were characterized with FTIR. Fig. 1a showed that there are obvious C-Br stretching vibration (582 cm^{-1}) in **TBMB** and C-Cl stretching vibration (675 cm^{-1}) in **BCMBP** and **BCMB**.[34] Also, the typical Si-O-Si stretching vibration at 1130 cm^{-1} was observed in **Ph-POSS**.[35] Th peaks of Si-O-Si in the polymers were broader than in the **Ph-POSS** monomer which was led by the hyper-networks. The stretching vibration at 2930 cm^{-1} was observed in **PPOSS** polymers that belonged to methylene groups. The disappearance of C-halogen stretching vibration peaks and the existence of Si-O-Si peaks (1130 cm^{-1}) and C–H vibration (2930 cm^{-1}) in the skeletons of the polymers proved the success of the polymerization.

The SEM images showed **PPOSS-1**, **−2**, **−3** were irregular blocks (Fig. S1) and no obvious sharp peaks in PXRD patterns also proved the amorphous phase of the **PPOSS** polymers (Fig. 1b). The irreversible Friedel-Crafts alkylation reaction led to the formation of the amorphous polymers.[36] The TEM images showed the polymers with irregular wormlike structures (Fig. S2). The thermal stability of the polymers was explored by TGA. From TGA plot, the **PPOSS** polymers showed great thermal stability and they started to decompose from 400 °C (Fig. 1c). After 600 °C, there were still 12%-20% residuals in the ashes which can be attributed to SiO_2. The chemical stability was also explored. The **PPOSS** polymers cannot dissolve in common organic solvents, such as ethanol, CH_2Cl_2, THF, DMF, etc. The strong covalent bonds and rigid building blocks made the **PPOSS** polymers with great stability.

The N_2 sorption was recorded at 77 K to characterize the porosity of

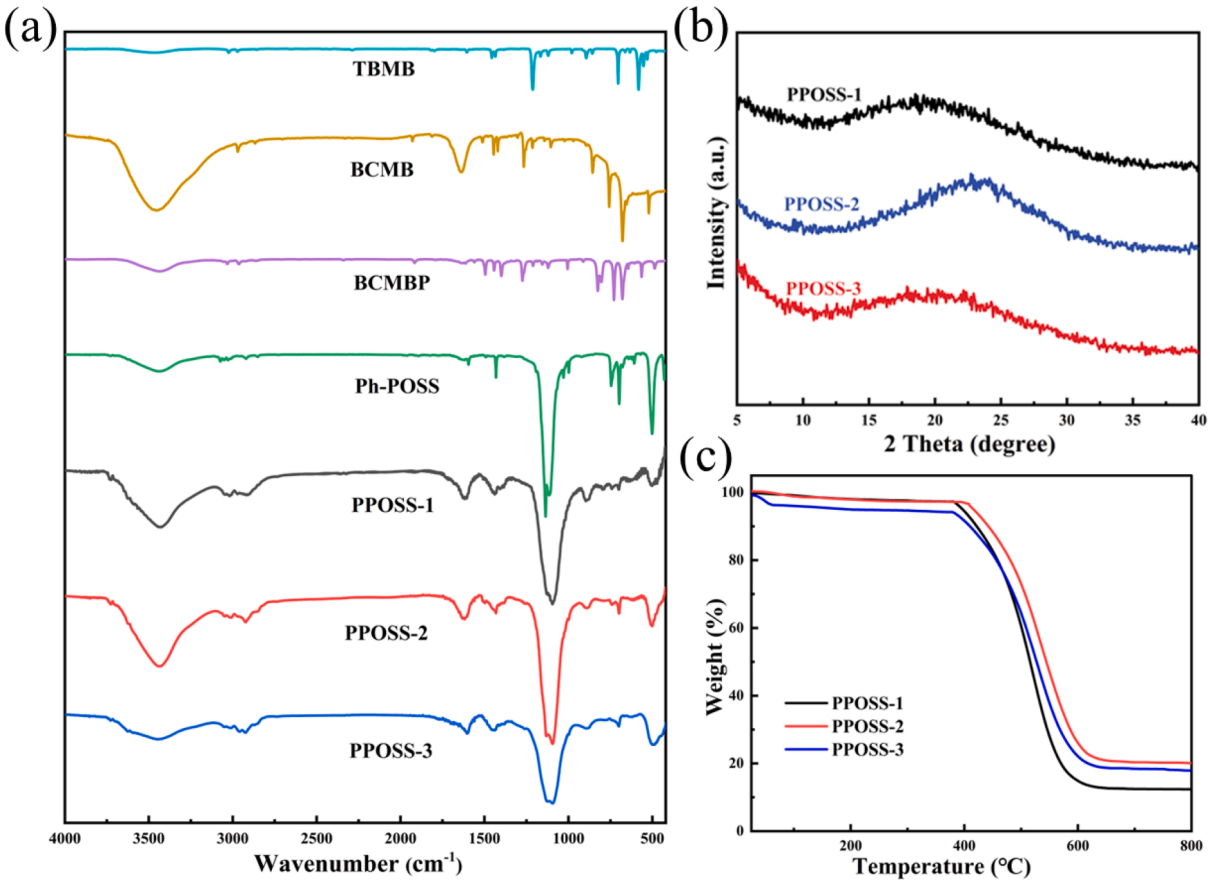


Fig. 1. (a) The FTIR of the monomers and synthesized **PPOSS** polymers; (b) PXRD patterns and (c)TGA plots of **PPOSS-1**, **−2**, **−3**.

Table 1
The BET surface area and pore volumes of **PPOSS-1**, **−2**, **−3**.

Sample	S_{BET} (m^2/g)	S_{micro} (m^2/g)	S_{micro}/S_{BET} (%)	V_{total} (cm^3 g^{-1})	V_{micro} (cm^3 g^{-1})	V_{micro}/V_{total} (%)
PPOSS-1	1113	838	75	1.4065	0.3657	26
PPOSS-2	973	589	60	1.0750	0.2579	24
PPOSS-3	1089	663	61	0.8993	0.2912	32

PPOSS polymers. The **PPOSS-1**, **−2**, **−3** exhibited similar adsorption process. There is a sharp increasement of adsorption at low relative pressure which indicated the existence of microporosity. Then with increase of the relative pressure, the adsorption capacity gradually raised (Fig. S3).[37] The large building block (**Ph-POSS**) and different connected linkers in the skeletons made the polymers possess great porosity. The BET surface areas of them were in the range of 973 m^2g^{-1} —1113 m^2g^{-1} (Table 1). All of them exhibited microporous and mesoporous composite pore size. Among them, **PPOSS-3** showed typical type IV isotherm with obvious hysteresis loop at high relative pressure. The pore width distribution was simulated by nonlocal density functional theory (NLDFT) and mainly distributed at 0.53 nm and 2.1 nm (Fig. S3). The percentage of microporous volume in total volume is 24% ~ 32% (Table 1).

Here, four PAHs, naphthalene (**NAP**), anthracene (**ANT**), acenaphthene (**AC**) and acenaphthylene (**ACE**) were chosen as the model aromatics that dissolved in *n*-cetane as model diesel. The **PPOSS-1**, **−2**, **−3** showed PAHs adsorption ability because of the high porosity and π-π interaction between polymers and PAHs. But due to the low microporous volume and pore size of the **Ph-POSS** polymers, the adsorption capacity was limited (Table S1).

To improve the microporosity and adsorption capacity, the **PPOSS-1**, **−2**, **−3** were carbonized directly at different high temperature (700 °C, 800 °C, 900 °C, and 1000 °C) to obtain PCMs. Then the PAHs adsorption capacity of **CPPOSS** PCMs was explored. The PCMs that carbonized at 1000 °C possessed highest PAHs adsorption capacity (Table S2). Among them, **CPPOSS-3–1000** showed excellent adsorption ability towards to four PAHs. Especially, the **NA** and **AN** adsorption can reach to 20.58 and 58.92 mg/g, respectively. The structure of **CPPOSS-3–1000** was characterized by solid-state ^{29}Si and ^{13}C NMR spectrum (Figure S4, S5). The chemical shift in ^{29}Si NMR spectrum at 103.0 ppm and 107.4 ppm demonstrate the presence of (Si (OSi)3OH) and (Si(OSi)4), respectively.[34] It proved the breakage of Si-C bonds after carbonization. The chemical shift in ^{13}C NMR spectrum at 122.1 ppm and 129.3 ppm indicated the existence of sp2 carbon atoms which illustrated the graphitization of the **CPPOSS-3–1000**.

To clarify the reason that PCMs possessed high PAHs adsorption capacity, the porosity of the PCMs was explored by N_2 sorption isotherms firstly. After carbonization, the micro- and mesoporous skeleton collapsed and varying degrees of decline of BET surface areas of the obtained **CPPOSS** PCMs occurred (Fig. 2, Table S3). The BET surface area of the **CPPOSS** increased with carbonized temperature because high temperature led to more porosity. Among the carbon materials obtained from different temperatures, **CPPOSS-1000** possessed the highest surface areas (719 m^2/g —842 m^2/g). By comparison with **PPOSS-1**, **−2**, **−3**, the total pore volumes of **CPPOSS-1000** decreased but there was no obvious reduction of microporous volume (Table S3). That means the decrease of BET surface area and pore volume after carbonization was mainly led by the reduction of mesopores. The micropore width of the carbonized PCMs reduced with the increase of

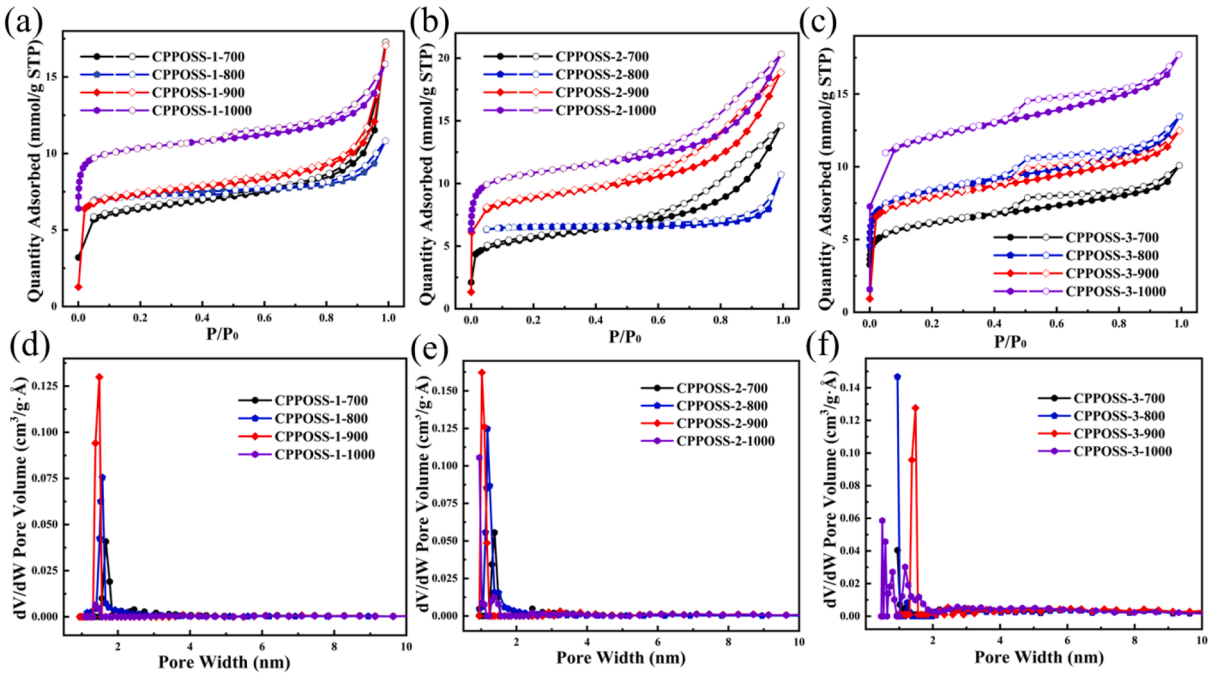


Fig. 2. (a) N_2 adsorption–desorption isotherms recorded at 77 K; (b) The pore size distributions of **CPPOSS-1**, **−2**, **−3** at different carbonization temperature.

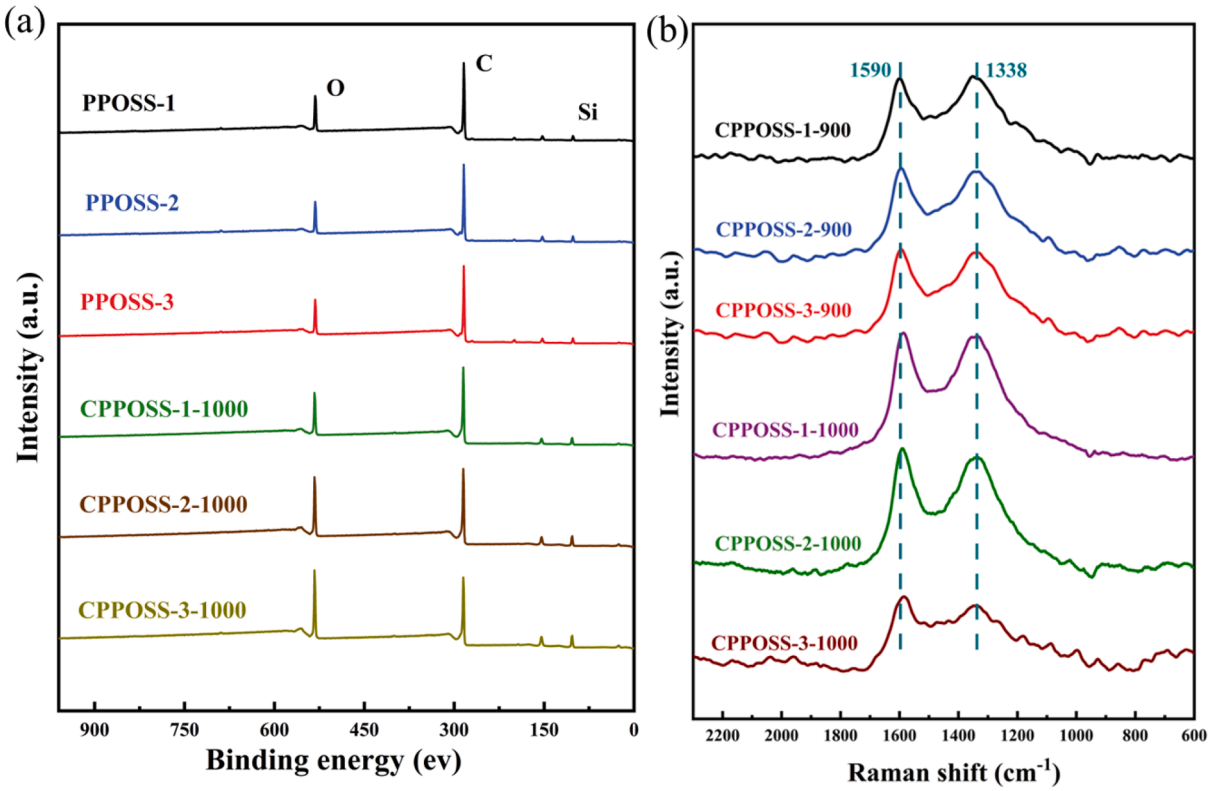


Fig. 3. (a) The XPS spectrum of **PPOSS-1**, **−2**, **−3** and **CPPOSS-1–1000**, **−2–1000**, **−3–1000**; (b) The Raman spectra of **CPPOSS-1–900/1000**, **−2–900/1000**, **−3–900/1000**.

carbonization temperature (Fig. 2d-e). **CPPOSS-3–1000** possessed more abundant hierarchical pore width (0.81 nm, 1.18 nm and 2.6 nm) than **CPPOSS-1–1000** (0.95 nm) and **CPPOSS-2–1000** (0.96 nm). The hierarchical pore size made PAHs enter into PCMs through the mesopores and be adsorbed in the skeletons efficiently.[38].

The more detailed characterization of **CPPOSS-1000** was carried out to indicate the high PAHs adsorption capacity. From the FTIR spectra, most peaks of exquisite functional groups in the carbonized materials (**CPPOSS-1000**) disappeared (Fig. S6). Slight Si-O-Si stretching vibration at 1130 cm^{-1} was observed because of the strong bond energy (460 kJ/mol). The breakage of most Si-O-Si bond exposed O atoms in the skeleton of PCMs. The XPS spectra also showed the relative intensity of

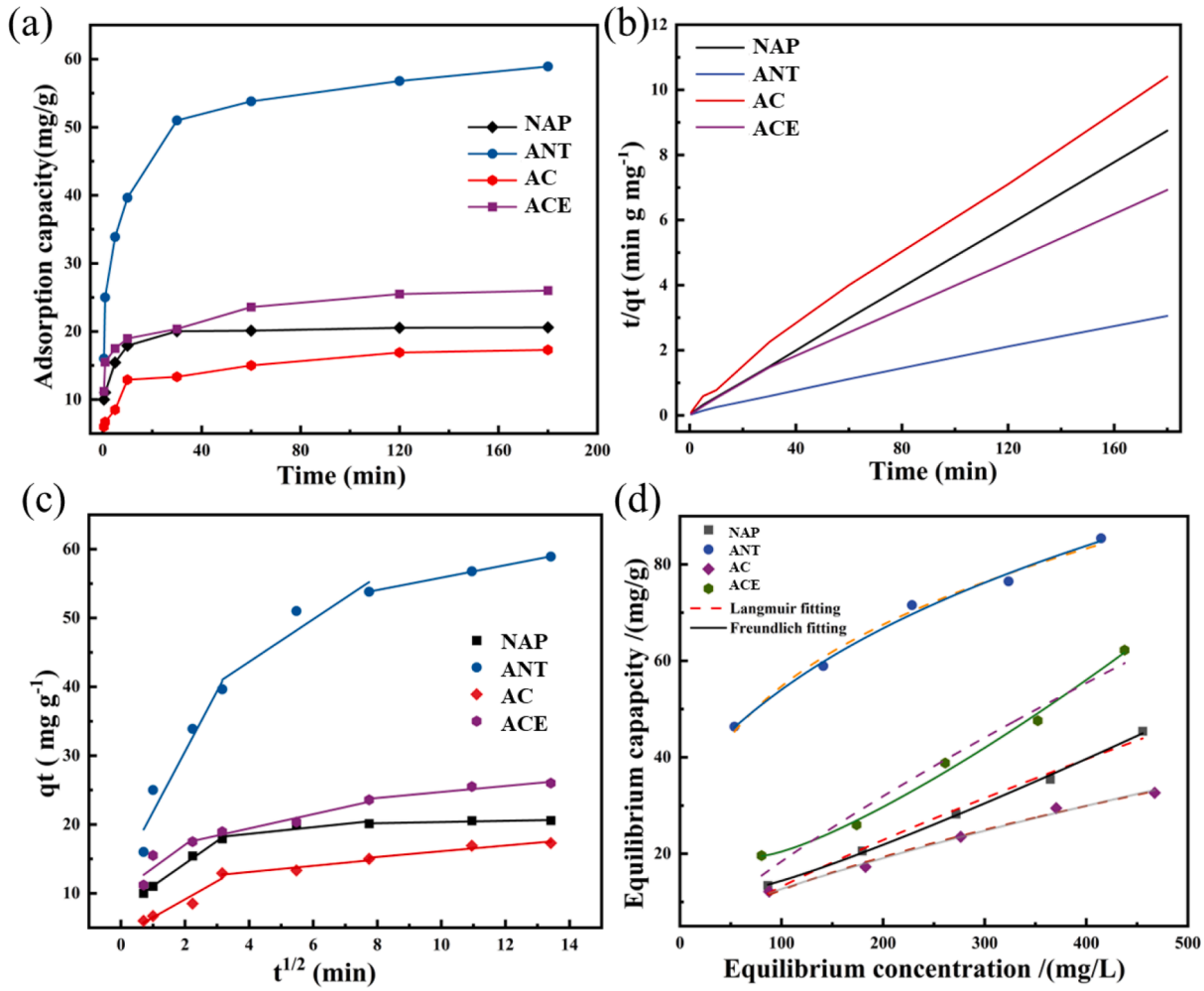


Fig. 4. The **NAP**, **ANT**, **AC** and **ACE** adsorption with **CPPOSS-3–1000**. (a) The kinetic curve of adsorption process; (b) The pseudo-second-order model fitted curves; (c) The intra-particle diffusion model; (d) Equilibrium adsorption isotherms. Langmuir and Freundlich models were showed with solid lines and dashed lines.

O and Si increased after carbonization (Fig. 3a). The increase of percentage of O in the PCMs can provide more p-π interaction site in the PAHs adsorption. The Raman spectrum showed two obvious peaks at 1590 cm^{-1} and 1338 cm^{-1} which were attributed to the graphite lattice (G band) and the disordered defects (D band). These two peaks indicated the carbonized materials were amorphous and with partially graphitization (Fig. 3b). The ratio of the intensity of D and G band (I_D/I_G) at different temperature was calculated. With the increasement of temperature, the value of I_D/I_G reduced which indicated the decrease of disorder and the increase of degree of graphitization (Table S4).[39] The peaks at 24° and 43° in XRD patterns of **CPPOSS-1000** which belong to (0 0 2) and (1 0 0) crystal planes of graphite also proved the happening of graphitization (Fig. S7).[38].

By combination of the above characterization results, the excellent PAHs adsorption ability of **CPPOSS-3–1000** can attribute to hierarchical pore size, high degree of graphitization and bare O atoms in the skeletons. The hierarchical pore size allowed more PAHs to enter into PCMs through larger micropores and adsorbed in the smaller micropores. The high degree of graphitization provided abundant π-π interaction and bare O atoms formed p-π conjugation between PCMs and PAHs. The synergistic effect achieved high effective PAHs adsorption.

The kinetic adsorption and equilibrium adsorption of **CPPOSS-3–1000** were also explored. Fig. 4a showed the quick adsorption of four PAHs happened in first 30 mins and reached equilibrium after 60 mins. The kinetic adsorption process was fitted with different model and the R^2 of pseudo-second-order model was more than 99% which indicated the chemical adsorption process (Fig. 4b, Table 2).[40] The value of K_2 can reveal the adsorption speed. It indicated that the adsorption speed of **NAP** was fastest and **ANT** was slowest. The adsorption speed of **AC** and **ACE** was similar. It may lead by the **NAP** possessed smallest molecular size (0.72 × 0.88 nm) in these four PAHs. **ANT** possessed longest molecular size (0.72 × 1.14 nm) and the size of **AC** (0.82 × 0.91 nm) and **ACE** (0.85 × 0.91 nm) was close. It was supposed that **ANT** molecules had to adjust the directions to enter into PCMs which prolonged the adsorption time and decreased adsorption speed. The intra-particle diffusion model also exhibited the quick adsorption in first 10 mins (Fig. 4c). The fitting cover did not pass through the origin which

Table 2
Adsorption kinetic parameter fitting for four PAHs adsorptions with **CPPOSS-3–1000**.

		NAP	ANT	AC	ACE
Pseudo-first-order	$q_{e,exp}$ (mg g^{-1})	20.58	58.92	17.30	26.00
	$q_{e,cal}$ (mg g^{-1})	6.0946	27.9026	9.3203	11.1987
	K_1 (min^{-1})	0.0439	0.0237	0.0260	0.0258
	R^2	0.9214	0.9023	0.9578	0.9801
Pseudo-second-order	$q_{e,cal}$ (mg g^{-1})	20.71	59.38	17.53	26.24
	K_2 ($g \cdot mg^{-1} \cdot min^{-1}$)	0.0398	0.0045	0.0120	0.0107
	R^2	0.9999	0.9991	0.9973	0.9981

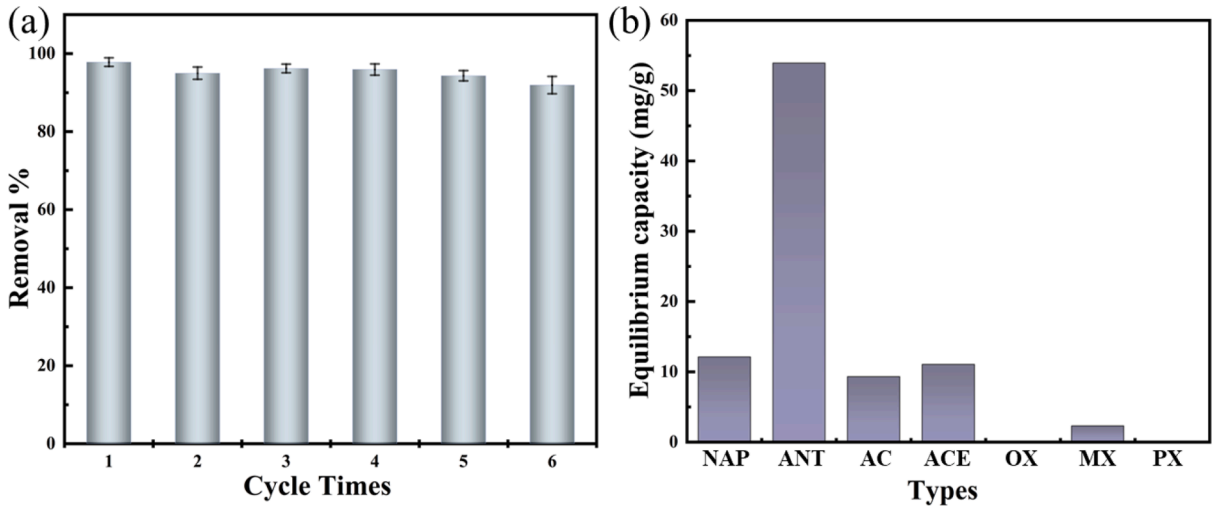


Fig. 5. (a) The recycling test of **ANT** adsorption with **CPPOSS-3–1000**; (b) The adsorption of four PAHs mixture in the existence of interference aromatics (o-xylene, oX; m-xylene, mX; p-xylene, pX).

indicated the adsorption speed of PAHs was not only depended on the diffusion rate in the PCMs (Table S5).[41] Here, it may also lead by the π-π and p-π interaction between the skeleton and PAHs which was proved by the FTIR and XPS spectrum (Figure S8, S9). In the FTIR spectra, the typical Si-O-Si stretching vibration at 1089 cm^{-1} and Si-C stretching vibration at 802 cm^{-1} were observed in **CPPOSS-3–1000**. After the adsorption of PAHs, the peak of Si-O-Si shifted to 1099 cm^{-1} and Si-C shifted to 810 cm^{-1}. The happening of shift of Si-O and Si-C proved the existence of π-π and p-π interaction between skeleton and PAHs. The O1s XPS spectrum of **CPPOSS-3–1000** after PAHs adsorption showed the shift from 532.6 eV to 532.9 eV which also proved the p-π interaction between O atoms and the conjugated system in PAHs (Figure S9).

The adsorption capacity of **CPPOSS-3–1000** were increased with the concentration (Fig. 4d). The adsorption data was fitted with Langmuir and Freundlich models. The simulated of correction coefficient of Freundlich model is higher than 0.99 (Table S6). It indicated that Freundlich model was more suitable for the equilibrium isotherm which proved the adsorption is a multi-layers adsorption process.[42].

The adsorption stability of **CPPOSS-3–1000** was explored by the cycle test (Fig. 5a). **CPPOSS-3–1000** can still keep high adsorption capacity and no obvious reduce of the removal ratio was observed after 6 cycles. The four PAHs were mixed together with three xylene isomers to test the anti-interference of **CPPOSS-3–1000**. There was a slight decrease of the adsorption capacity of four PAHs compared with the single adsorption results in existence of xylenes (Fig. 5b). But the total PAHs adsorption capacity was high. For the monocyclic aromatic, xylene isomers, the adsorption capacity was low (<5 mg/g) which proved the good selective adsorption of PAHs.

4. Conclusion

In this study, **Ph-POSS** was introduced into the synthesis of POPs under mild conditions. The synthesized POPs possessed high BET surface area (~1000 m^2/g) and narrow micropores. The corresponding PCMs were obtained by carbonization at different high temperatures and the porosities were well characterized. The porosity was maintained and micropore width was increased. The suitable pore size enhanced the PAHs adsorption ability. Furthermore, the graphitization that caused by carbonization formed π-π interaction with PAHs. The bare O atoms possessed lone electric pairs and provided p-π conjugation. **CPPOSS-3–1000** exhibited highest PAHs adsorption capacity and good cycle performance. The anti-interference in the xylene isomers made as a potential candidate of PAHs adsorbents in chemical industry.

CRediT authorship contribution statement

Fangfang Zhang: Conceptualization, Investigation, Writing – original draft. **Peng Cui:** Conceptualization, Methodology, Writing – review & editing, Supervision. **Dongni Liu:** Validation, Investigation. **Zhuo Han:** Formal analysis, Methodology. **Jie Zhu:** Visualization. **Yan Huang:** Data curation. **Yanhong Chao:** Methodology. **Yuanbin She:** Formal analysis. **Wenshuai Zhu:** Conceptualization, Writing – review & editing, Project administration, Funding acquisition, Supervision. **Zhichang Liu:** Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2023.112583.

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